1.Data Modelling and Cleaning

- **Removed duplicate rows from Customer_ID column in Customer_Demographics Table.**
- **Formatted the columns of the all the tables to their accurate data types. United Kingdom date format for all the date columns, and correct numeric data types for numeric columns such as decimal for Amount, whole number Points_Earned, Points_Redeemed, etc.**

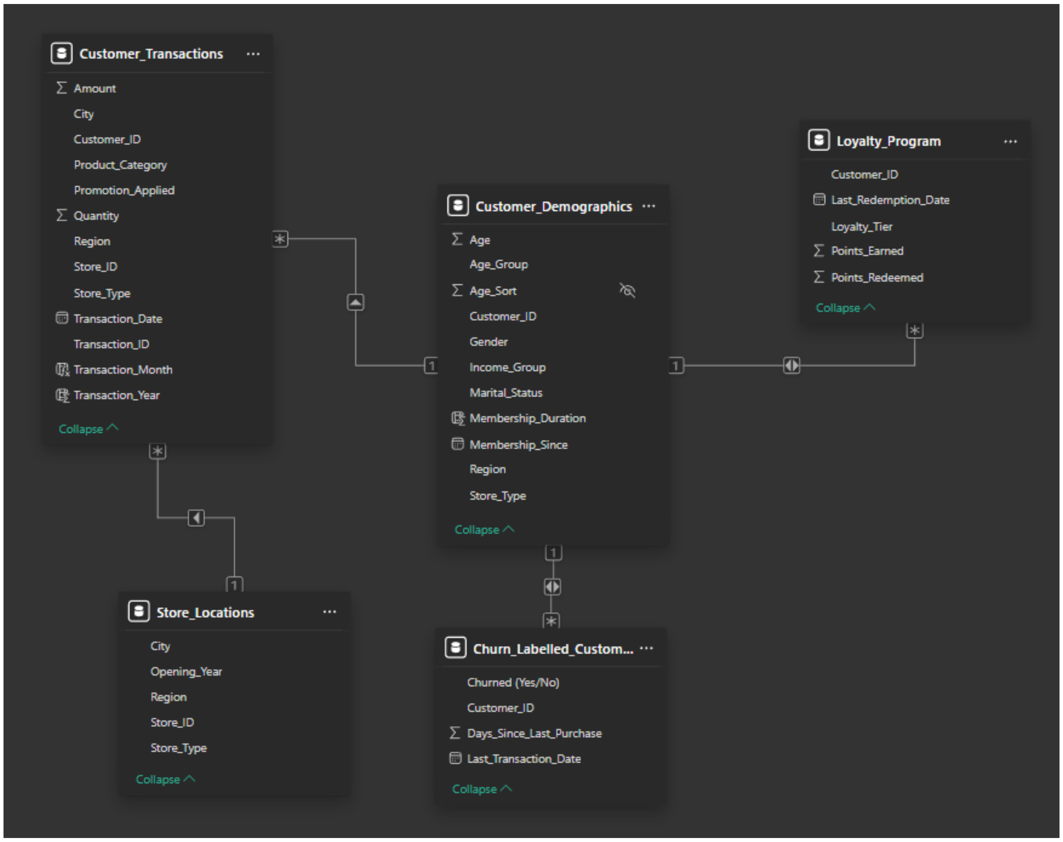
Customer_ID	Gender	Age	Membership_Since	Marital_Status	Region	Income_Group	Membership_Duration
C1000	Male	50	01 November 2020	Single	London	High	6
C1002	Male	36	18 August 2021	Single	Birmingham	Medium	5
C1005	Female	61	19 November 2020	Single	Liverpool	High	6
C1010	Male	21	03 May 2023	Single	Manchester	Medium	3
C1017	Female	51	11 July 2023	Single	London	High	3
C1018	Female	23	21 March 2024	Single	Liverpool	Low	2
C1019	Male	39	07 June 2023	Single	Leeds	Low	3
C1022	Female	33	19 December 2022	Single	London	Low	4
C1026	Female	33	24 November 2023	Single	Birmingham	Medium	3
C1027	Female	46	08 September 2021	Single	Birmingham	High	5
C1028	Female	20	30 August 2021	Single	Liverpool	Low	5
C1031	Male	36	26 November 2023	Single	Manchester	Medium	3
C1033	Female	20	02 August 2022	Single	Birmingham	High	4
C1034	Female	36	26 August 2023	Single	Birmingham	Low	3
C1037	Male	24	05 April 2023	Single	London	Low	3
C1038	Male	69	10 February 2024	Single	London	Medium	2
C1042	Female	56	26 February 2024	Single	London	Low	2
C1047	Male	45	08 October 2023	Single	London	High	3
C1049	Female	67	14 September 2023	Single	London	Low	3
C1056	Female	33	10 September 2021	Single	Leeds	Low	5
C1059	Male	69	30 June 2022	Single	Birmingham	High	4
C1061	Male	19	29 June 2021	Single	London	Low	5
C1063	Male	65	26 December 2022	Single	Liverpool	High	4
C1064	Male	29	07 September 2023	Single	Liverpool	Medium	3
C1065	Female	22	01 October 2022	Single	London	Medium	4
C1066	Female	54	04 August 2020	Single	Birmingham	High	6
C1067	Male	49	10 May 2024	Single	Birmingham	Medium	2

^ Created a Calculated Column for Membership_Duration in Customer_Demographics table using the formula :
Membership_Duration = DATEDIFF(Customer_Demographics[Membership_Since], TODAY(), YEAR)

Customer_ID	Transaction_Date	Store_ID	Product_Category	Amount	Quantity	Promotion_Applied	Transaction_Month	Transaction_Year
C1011	11 November 2024	S106	Bakery	249.6	7	Yes	November	2024
C1080	05 July 2024	S106	Grocery	474.85	3	No	July	2024
C1013	01 September 2024	S106	Electronics	321.32	1	Yes	September	2024
C1187	21 October 2024	S106	Grocery	417.73	7	No	October	2024

^ Created a Calculated Column for Transaction_Month and Transaction_Year from Transaction_Date Column in Customer_Transactions table using the following formulas :
Transaction_Month = Customer_Transactions[Transaction_Date].[Month] and Transaction_Year = Customer_Transactions[Transaction_Date].[Year]

(Note : All the measures and formulas are mentioned in Italics.)

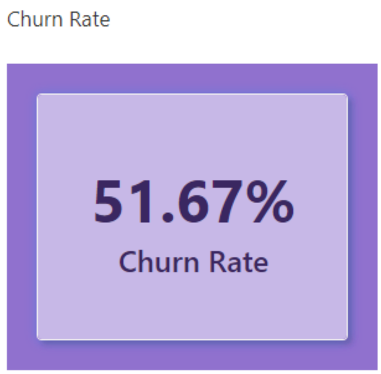


^ Created the following Data Models using defined relationships :

- **One-to-Many (1 → *) : Customer_Demographics -> Customer_Transactions, Loyalty_Program and Churn_Labelled_Customers using Customer_ID for all the relationships. Customer_Demographics is the Fact Table and the others are Dimension Tables and together they form a Fact-Dimension Table or a Star Schema.**
- **Many-to-One (* → 1) : Customer_Transactions -> Store_Locations using Store_ID.**

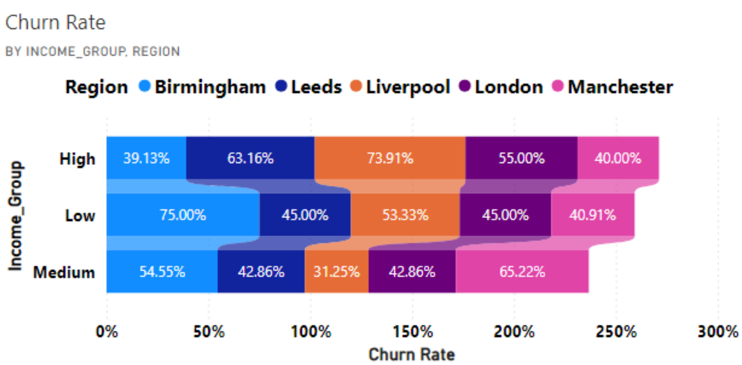
(Note : The Reports and respective Visualizations will be displayed from the next page.)

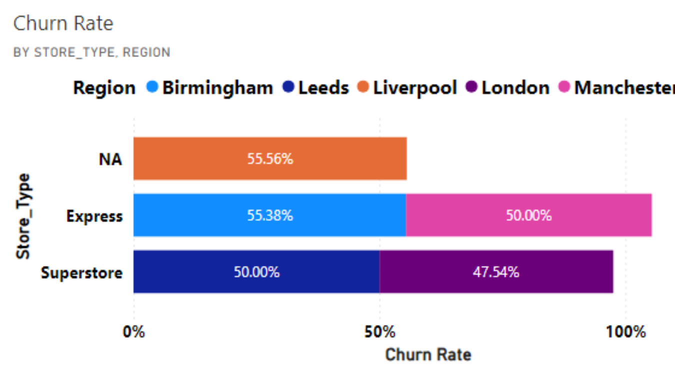
2.Churn And Retention Metrics



Created a Churn Rate Card visual using the following formulas/measures :

- *Total Customers = DISTINCTCOUNT(Customer_Demographics[Customer_ID])* -> to calculate the total number of customers.
- *Churned Customers = CALCULATE(DISTINCTCOUNT(Churn_Labelled_Customers[Customer_ID]),Churn_Labelled_Customers[Churned (Yes/No)] = "Yes")* -> to calculate the number of churned customers.
- *Churn Rate = DIVIDE([Churned Customers],[Total Customers])* -> to calculate the churn rate.



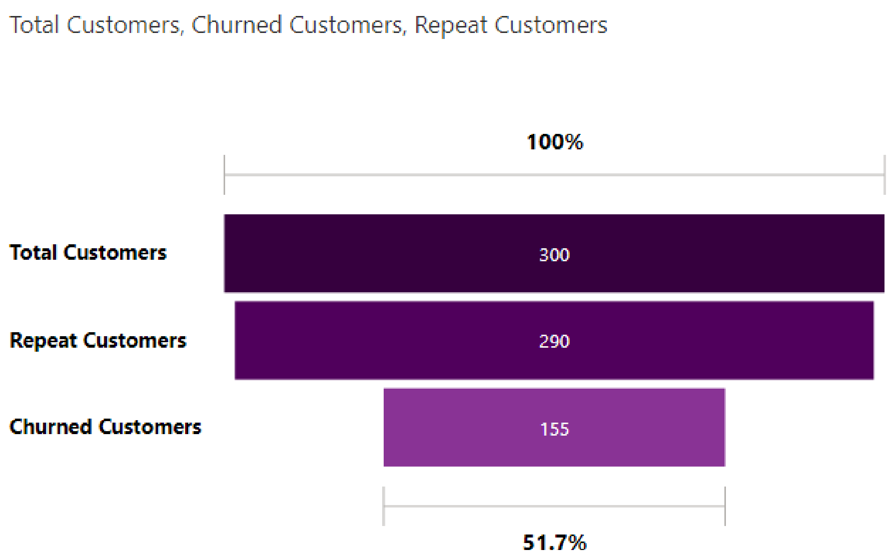


^ 1st Hierarchy ^ 2nd Hierarchy

Created a Stacked Bar chart for visualizing Churn Rate by Income_Group, Region and Store_Type with the two respective displayable hierarchies which are mentioned above. For getting the visualization for Store_Type, I merged queries in the Power Query editor, merging Customer_Demographics with Store_Locations using Store_ID column (replaced the blank values in Store_Type for Liverpool region with 'NA' after merging). And for further clarity, I used a Table visual given below for visualizing the same.

Churn Rate	Region	Income_Group	Store_Type
75.00%	Birmingham	Low	Express
73.91%	Liverpool	High	NA
65.22%	Manchester	Medium	Express
63.16%	Leeds	High	Superstore
55.00%	London	High	Superstore
54.55%	Birmingham	Medium	Express
53.33%	Liverpool	Low	NA
45.00%	Leeds	Low	Superstore
45.00%	London	Low	Superstore
42.86%	Leeds	Medium	Superstore
42.86%	London	Medium	Superstore
40.91%	Manchester	Low	Express
40.00%	Manchester	High	Express
39.13%	Birmingham	High	Express
31.25%	Liverpool	Medium	NA
51.67%			

- **From the table visual in previous page we can easily identify the top 5 segments with the highest Churn Rate i.e., which 5 income groups from which 5 regions with which type of store (2 types and 'NA' for Liverpool) has the highest churn rates. I have sorted the Churn Rate column in descending order and used data bars on its cells. For example, we can see that the Low Income group from Birmingham region with Express stores has had the Highest Churn Rate i.e., 75%.**

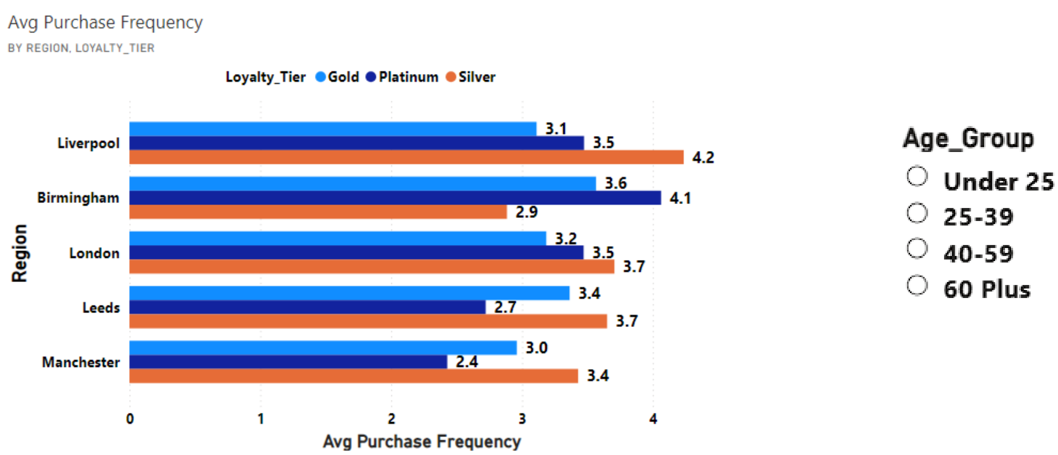


^ Used a Funnel chart to visualize the progressive decrease in the number of customers through the stages : Total Customers → Repeat Customer → Churned Customers. I already had measures for Total and Churned customers, but created a new measure for Repeat customers :

Repeat Customers = COUNTROWS(FILTER(VALUES(Customer_Transactions[Customer_ID]), COUNTROWS(Customer_Transactions) > 1))

3.Repeat Purchase Analysis

- Created DAX Measures for the following :
  1. Low-Tier Customers: (2-4 purchases) -> *Low-Tier Customers = CALCULATE(DISTINCTCOUNT(Customer_Transactions[Customer_ID]), FILTER(VALUES(Customer_Transactions[Customer_ID]),[Total Customers] >= 2 && [Total Customers] <= 4))*
  2. Mid-Tier Customers: (5-10 purchases) -> *Mid-Tier Customers = CALCULATE(DISTINCTCOUNT(Customer_Transactions[Customer_ID]), FILTER(VALUES(Customer_Transactions[Customer_ID]),[Total Customers] >= 5 && [Total Customers] <= 10))*
  3. High-Tier Customers: (11+ purchases) -> *High-Tier Customers = CALCULATE(DISTINCTCOUNT(Customer_Transactions[Customer_ID]), FILTER(VALUES(Customer_Transactions[Customer_ID]),[Total Customers] >= 11))*



^ Used a Clustered Bar chart for visualizing Average Purchase Frequency by Region (from Customer_Demographics table) and Loyalty tier using the three respective loyalty tiers (from the Loyalty_Program table) as legend with an interactive slicer for Age group, so it displays the corresponding comparison of the metrics mentioned in the chart for each individual age group when one is selected. I've created a custom conditional column for the Age groups (Age_Group) from the ages given in the Age column in Customer_Demographics table and another conditional column for sorting the age groups in ascending order (Age_Sort) which I have hidden in the report view. I have used a measure for Average Purchase Frequency :
Avg Purchase Frequency = DIVIDE(COUNT(Customer_Transactions[Customer_ID]),[Total Customers])

Repeat Customers
BY PRODUCT_CATEGORY

Repeat Customers	Product_Category
149	Clothing
145	Grocery
141	Beverages
140	Bakery
136	Electronics
290	

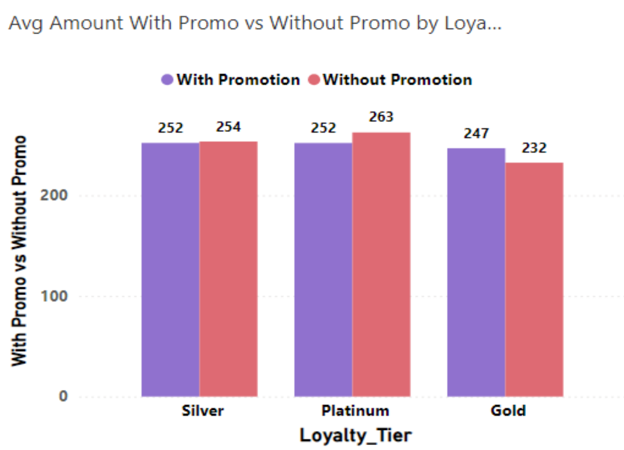
^ I have used a table matrix given above for displaying the most frequent categories of product bought by loyal customer by using Repeat Customers column (a measure mentioned in the previous task .i.e., Churn And Retention Metrics) and Product_Category column (from Customer_Transactions table). I have sorted the number of Repeat Customers in descending order and then, we can see that Clothing is the most frequently bought product category with 149 repeat customers.

4.Promotion And Loyalty Impact

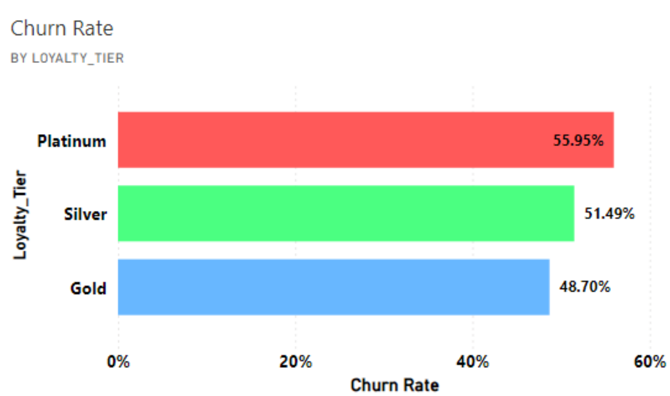
- **Created the following Measures :**

1. ***% Transactions with Promotion = DIVIDE(CALCULATE(COUNT(Customer_Transactions[Transaction_ID]),Customer_Transactions[Promotion_Applied] = "Yes"),COUNT(Customer_Transactions[Transaction_ID]),0)***
   (I have not visualized ^ this section).
2. **Average purchase amount with vs without promotion (in 2 parts respectively) ->**
   - ***Avg Purchase Amount With Promotion = CALCULATE(AVERAGE(Customer_Transactions[Amount]),Customer_Transactions[Promotion_Applied] = "Yes")*** **&**
   - ***Avg Purchase Amount Without Promotion = CALCULATE(AVERAGE(Customer_Transactions[Amount]),Customer_Transactions[Promotion_Applied] = "No")***

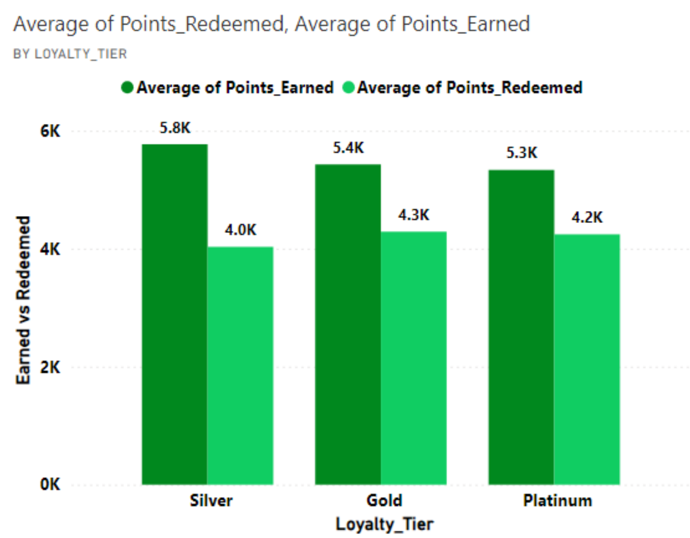
I have used the following Clustered Column chart to visualize Average purchase amount with promotion and the same without promotion against the 3 respective loyalty tiers i.e., Silver, Gold and Platinum.



- **I have used the following Bar chart to compare the churn rate across the 3 different loyalty tiers.**



- **The Bar chart above shows that the Platinum loyalty tier has the highest Churn rate i.e., 55.95%.**



^ Used a Clustered Column chart to compare the Average points earned (from Points_Earned column from the Loyalty_Program table) and Average points redeemed (from Points_Redeemed column from the Loyalty_Program table as well) across the loyalty tiers. I have also used the respective averages as legend for clearly understanding the colour coordination of the clustered columns.

← Used Region from Customer_Demographics table as an universal slicer for all the charts and visuals in this task i.e., Promotion and Loyalty Impact to compare the respective metrics across each individual region at a time.

Recommendations to improve Redemption and Retention :

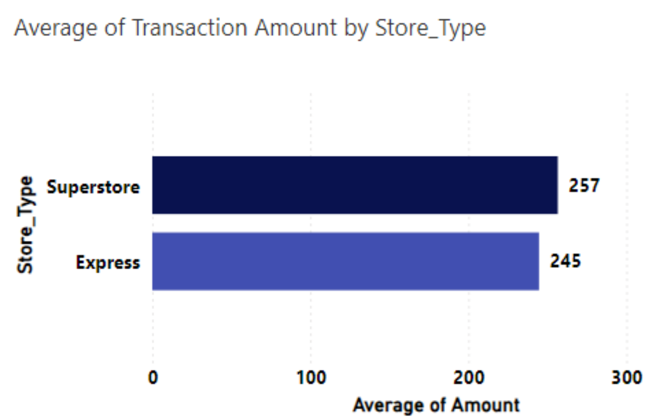
IKEA can track their loyalty program closely and offer all kinds of desirable rewards such as discounts and sale (once or twice a month) on electronics especially (as it is the least frequently bought product category by loyal customers). In general, they can communicate with their customers via multiple channels such as mail, SMS or app about point balances and implement expiration dates, create engaging content for various age groups, provides personalized gifts and exception customer service.

(Note : All Recommendations and Suggestions are given in text boxes filled with yellow colour.)

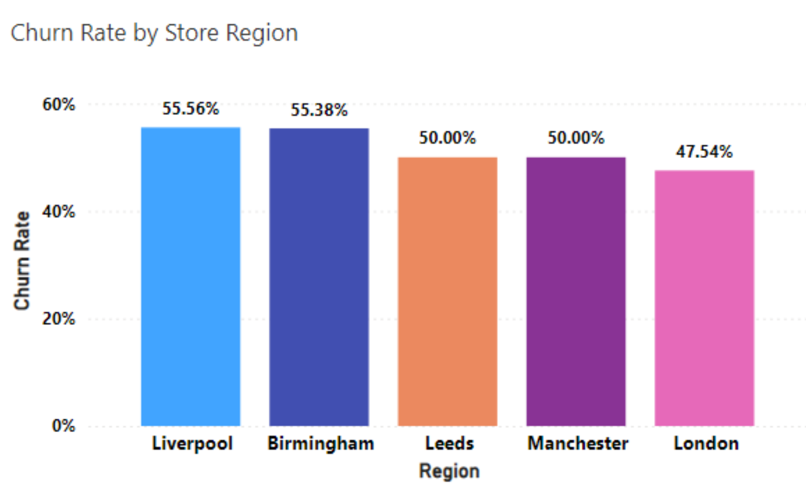
5.Store Performance vs Retention

- Used Merge Queries in Power Query editor to join Customer_Transactions table with Store_Locations table using Store_ID column.
- Visualized the following :

1. Average transaction amount (which is the average of the Amount column in the Customer_Transactions table) by the 2 respective types of store (Store_Type column from Customer_Transactions table after merging the queries as previously mentioned) in the following bar chart :



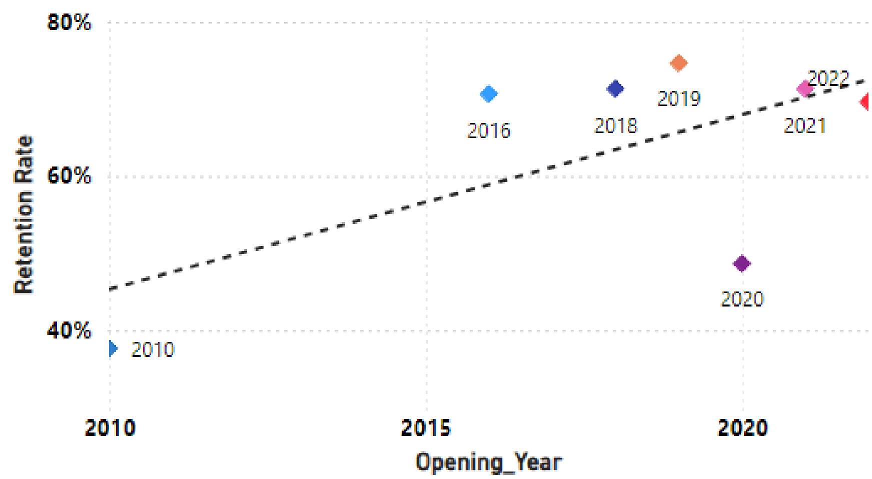
2. Churn rate by store region (Region column from Customer_Demographics table) using the following column chart :



From the column chart above, we can clearly identify that the stores in Liverpool region have the highest Churn rate i.e., 55.56%.

3. Correlation between store opening year (Opening_Year from Store_Locations table) and Retention (Retention Rate, which I have calculated using a custom measure i.e., *Retention Rate = 1 - DIVIDE([Repeat Customers], [Total Customers])*) using the Scatter chart given below :

Correlation between Store Opening Year And Retention



^ From the scatter chart above, we can clearly see that the store which opened in the year 2019 has the highest Retention rate.

After a thorough analysis of the IKEA store locations and how they are affected by churning, I'd suggest the organization to run store-specific public engagement campaigns in Liverpool which has the highest churn rate and in Birmingham which has the second highest churn rate.

6.Customer Value (CLV) Analysis

- **Created a custom DAX Measure for calculating CLV :**

CLV =
DIVIDE(SUM(Customer_Transactions[Amount]),SUM('Customer_Demographics'[Membership_Duration]))

Divided the customers into 3 respective segments -> Low (Bottom 25%), Medium (Mid 50%) and High (Top 25%) based on their CLVs using :

1. **A Custom DAX Segment Datatable which stores each individual segment as a string :-**

```
CLV Segment Table =
DATATABLE (
  "Segment", STRING,
  {
    {"Low (Bottom 25%)"},
    {"Medium (Middle 50%)"},
    {"High (Top 25%)"}
  }
)
```

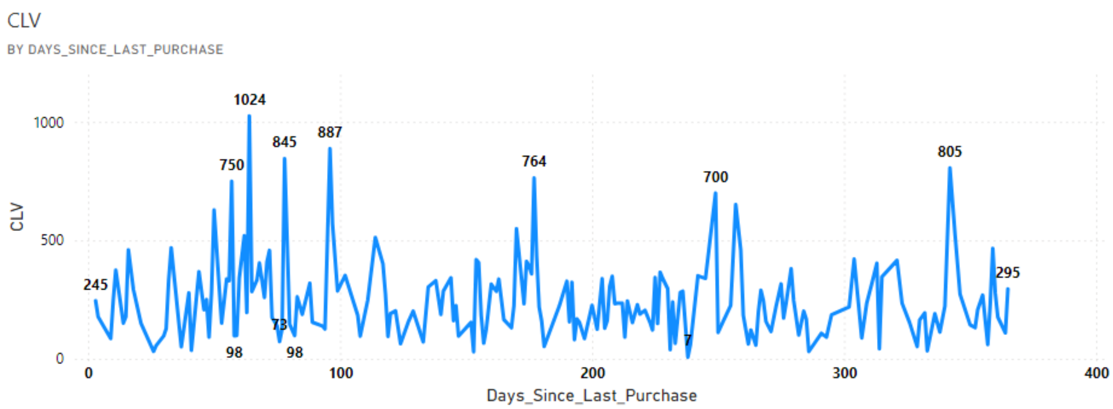
2. **A Custom logical DAX measure which checks the logical conditions corresponding to the respective segments in the datatable and gives a segment as an output if the aligning condition is satisfied :**

```
CLV Segment Logic =
VAR CustomerCLV = [CLV]

VAR P25 =
  PERCENTILEX.INC (
    ALL ( Customer_Demographics[Customer_ID] ),
    [CLV],
    0.25
  )

VAR P75 =
  PERCENTILEX.INC (
    ALL ( Customer_Demographics[Customer_ID] ),
    [CLV],
    0.75
  )

VAR CustomerSegment =
  SWITCH (
    TRUE (),
    CustomerCLV <= P25, "Low (Bottom 25%)",
    CustomerCLV <= P75, "Medium (Middle 50%)",
    "High (Top 25%)"
  )
```

```
VAR SelectedSegment =
  SELECTEDVALUE ( 'CLV Segment Table'[Segment] )

RETURN
IF (
  ISBLANK ( SelectedSegment )
    || SelectedSegment = CustomerSegment,
  1,
  0
)
```

- **Visualized the following :**

1. **CLV vs Days Since Last Purchase (from Churn_Labelled_Customers table) using a Line chart :**



^ From the above given line chart, it's fair to say that the majority of customers with High CLV or High-Value customers are at risk of churning.

2. **CLV by Loyalty Tier and Region (from Customer_Demographics table) using a table visual (I have taken Loyalty_Tier as a slicer so, we can compare the CLVs for the respective regions across each individual tier. The region slicer also affects the line chart above so we can analyse it further based on a tier at a time) :**

CLV

BY REGION

CLV	Region
221	Birmingham
215	Leeds
239	Liverpool
221	London
196	Manchester
218	

Loyalty_Tier

- Gold
- Platinum
- Silver

7.Final Dashboard and Executive summary

^ This is my Final Dashboard that I created using Power BI online service features by pinning the most necessary visualizations of the most impactful metrics from my report and I have also attached it to the final page of the PDF file which contains the multi-page report.

8.Video Explaination

Drivelink for Explanation Video :

**https://drive.google.com/file/d/1kq6OY2WCRjYegN66khM0RzM5D
CIenNQU/view?usp=sharing**